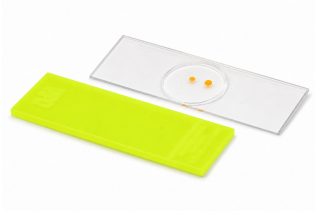
Quick Start Guide
FLIM-Phasors Calibration Kit

CAL-PH-K-00



1 About This Guide

This document is a condensed quick-start guide for basic handling and use of the FLIM-Phasors Calibration Kit. This guide does **not** replace the full product manual. The complete user manual can be downloaded at: https://flim-labs.github.io/docs/calibkit/Calibkit-manual.pdf

Scan the QR code to open the full manual.

Read the complete manual before using the kit, especially the handling, storage, and safety instructions.

2 Safety Information

The calibration kit contains glass and polymer reference slides. The product is not classified as hazardous, but improper handling may damage the samples or cause injury.

- Handle all slides carefully.
- Do not use damaged, cracked, or chipped slides.
- Broken glass can cause cuts.
- Use protective gloves when handling broken slides.
- Keep the kit out of reach of children.
- Store the kit in a clean and dry environment.

3 Intended Use

The FLIM-Phasors Calibration Kit is intended for calibration, verification, and qualitative assessment of FLIM and FLIM-Phasors systems in laboratory and research environments.

The kit is intended for research use only (RUO). It must not be used for medical, diagnostic, or clinical purposes.

4 Product Overview

The FLIM-Phasors Calibration Kit contains reference samples for checking fluorescence lifetime measurements and imaging performance.

The kit supports:

- verification of fluorescence lifetime accuracy
- generation of calibration data
- validation of phasor-plot positioning
- qualitative assessment of image quality
- training and demonstration workflows

5 Scope of Delivery

- Fluorescence lifetime reference sample
- Biological imaging reference sample
- Product documentation

6 Unpacking

The kit is supplied in protective packaging. Handle the slides carefully during unpacking.

After unpacking:

- inspect the slides for cracks, chips, or visible damage
- check that the optical surfaces are clean and free from scratches
- verify that the documentation is present

Broken glass can cause cuts or injury. Handle damaged samples with care and dispose of them according to local regulations.

7 Requirements for Use

Before using the kit, ensure that:

- the microscope or FLIM system is installed and operational
- the required excitation wavelength is available
- detector and acquisition settings are suitable for the selected sample
- FLIM LABS software is installed and configured
- environmental conditions are suitable for laboratory use

8 Principle of Use

The lifetime reference sample has a known mono-exponential fluorescence decay and can be used to verify lifetime accuracy or generate calibration data.

The biological imaging reference sample provides structured fluorescent features for qualitative assessment of image quality and phasor-space behavior.

For reliable calibration, the count rate during calibration should match the experiment. If necessary, reduce the excitation intensity using neutral density filters or by de-focusing the excitation beam.

Avoid touching the optical surface of the slide. Contamination, scratches, or damage may affect the measurement result.

9 FLIM-Phasors Calibration Procedure

General workflow:

1. Select the lifetime reference slide required for the measurement.
2. Place the slide into the microscope sample holder.
3. Ensure that the slide is seated securely.
4. Focus on the measurement area.
5. Adjust illumination intensity and detector settings to match the count rate of your experiential sample.
6. Start the calibration or verification workflow in the software as described in the following sections.

10 Using *SpectroscoPy* Software

1. Follow the general procedure for placing the lifetime reference slide and adjusting the optical setup.
2. Open *SpectroscoPy*.
3. Switch to *Spectroscopy Mode*.
4. Select *Phasors Ref* as the calibration type.
5. Enter the known decay constant in Tau (ns) .
6. Select the acquisition settings.
7. Acquire and export the reference data.

11 Using *FLIM IMAGER* Software

1. Follow the general procedure for placing the lifetime reference slide and adjusting the optical setup.
2. Open *FLIM IMAGER*.
3. Select the *Calibration* tab in the FLIM analysis modes.
4. Enter the known decay constant in Tau (ns) .
5. Start the calibration acquisition.
6. Apply the calibration data in *Phasors* mode.

For detailed instructions, refer to the *FLIM IMAGER* software manual: https://flim-labs.github.io/docs/imager/Imager-manual.pdf

12 Further Information

Additional information and downloads are available on the product website: https://www.flimlabs.com/products/phasors-calibration-kit/

13 Product Label

FLIM-Phasors Calibration Kit

P/N: CAL-PH-K-00
S/N: CAL-XXXX-XXX



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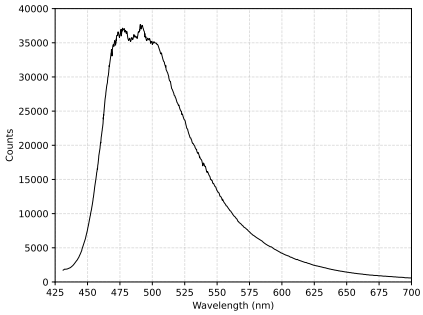


Figure 1: Emission spectrum of the lifetime reference sample when excited at 405 nm.

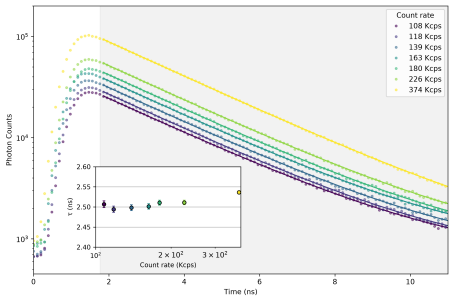


Figure 2: Measured lifetime τ of the lifetime reference sample by fitting TCSPC data acquired at different count rates (CPS), tuned via the 488 nm excitation laser power.

14 Technical Specifications

Parameter	Value
General	
Product name	FLIM-Phasors Calibration Kit
Product code	FLIM-CALB-00
Use classification	Research Use Only (RUO)
Intended environment	Indoor laboratory use
Lifetime Reference Sample	
Sample type	Fluorescent lifetime reference sample
Dimensions	75 mm x 25 mm x 3 mm
Weight	< 10 g
Material	PMMA
Fluorescence decay	Monoexponential
Reference lifetime	2.5 ns
Excitation wavelengths	405 nm – 488 nm
Emission spectrum at 405 nm	Peak: 495 nm FWHM: 77 nm Spectral centroid: 512 nm
Biological Imaging Reference Sample	
Sample type	Biological imaging reference sample
Dimensions	75 mm x 25 mm x 1 mm
Weight	< 10 g
Material	Glass
Biological specimen	*Convallaria majalis*
Environmental Conditions	
Operating temperature	0 °C to +40 °C
Storage temperature	-20 °C to +70 °C
Relative humidity	Up to 80% non-condensing
Storage conditions	Clean and dry environment at room temperature
Special precaution	Avoid prolonged exposure to intense light when not in use

Table 1: Key specifications of the FLIM-Phasors Calibration Kit.

15 Troubleshooting

Issue	Possible Cause	Recommended Action
Unexpected lifetime value	Incorrect software settings or calibration parameters	Verify acquisition settings, synchronization settings, and entered reference lifetime.
Weak fluorescence signal	Insufficient excitation intensity or incorrect focus	Check focus and excitation path. Increase illumination only within safe detector limits.
Detector saturation	Excitation intensity too high	Reduce excitation intensity using ND filters or defocus the beam.
Poor image quality	Incorrect focus, dirty slide, or unsuitable imaging settings	Refocus the sample, inspect the slide surface, and verify microscope settings.
Inconsistent calibration results	Different settings between reference and sample measurement	Use matching illumination, detector, acquisition, and reconstruction settings.
Visible scratches, cracks, or chips	Mechanical damage to the slide	Do not use the damaged slide. Handle broken glass carefully and dispose of it according to local regulations.

Table 2: Common issues when using the calibration kit.